

Mock Exam (50 Marks)

1. HTML5 Semantic Elements (10 marks)

- a) What are semantic elements in HTML5? Explain why they are important for web development. Provide at least 3 examples of semantic elements. (5 marks)
- b) Write a simple HTML5 structure using at least 3 semantic elements to define the header, main content, and footer of a web page. (5 marks)

2. Responsive Design (10 marks)

- a) Explain the concept of responsive design. Why is it essential in modern web development? (4 marks)
- b) Name and describe two elements/attributes that are important for building responsive websites. (4 marks)
- c) What is a media query in CSS, and how is it used for responsive design? Provide a basic example. (2 marks)

3. Flexbox Directions (10 marks)

- a) In Flexbox, what are the `flex-direction`, `wrap`, `order`, `basis` properties? Explain how each affects the layout of the flex items. (5 marks)
- b) Given a flex container, write CSS to display its child elements in a column layout and align them in the center both horizontally and vertically. (5 marks)

4. Specificity Rules (10 marks)

- a) Explain how CSS specificity works. Provide an example of how specificity affects the application of styles in a web page. (5 marks)
- b) Calculate the specificity of the following selectors and explain which one has higher specificity:
 - i) `.header .nav-item`
 - ii) `#main-content .article h1`
 - iii) `div p`(5 marks)

5. Simple Gradients (10 marks)

- a) What is a gradient in CSS? Explain the difference between a linear gradient and a radial gradient. (4 marks)
- b) Write a CSS rule to create a background that transitions from blue to green using a linear gradient. (3 marks)
- c) How can you control the direction of a linear gradient? Modify your answer in (b) to create a diagonal gradient. (3 marks)